

DEFLATING THE HESSIAN: RANK-4 W4A4 QUANTIZATION FOR MULTIMODAL DIFFUSION TRANSFORMERS

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ABSTRACT

In diffusion transformers, low-rank branches can mitigate 4-bit weight–activation (W4A4) post-training quantization (PTQ) loss by decomposing each weight into a low-bit residual and a high-precision low-rank component. Existing low-rank PTQ approaches, however, either optimize low-rank compensation and residual quantization separately, often requiring higher ranks, or rely on second-order weight updates without explicitly modeling activation quantization error, which becomes particularly pronounced under 4-bit quantization. To address these limitations, we present *H-SVDQuant*, a unified framework modeling low-rank-assisted W4A4 PTQ as a coupled calibration problem and deriving optimization-based solvers from the joint objective. Eliminating the output-side low-rank factor yields a *deflated Hessian* that discounts residual errors already captured by the low-rank component, while an activation-noise surrogate is incorporated to suppress activation quantization error. Across five diffusion backbones, rank-4 *H-SVDQuant* consistently outperforms rank-4 SVDQuant in PSNR and LPIPS. It further surpasses rank-32 SVDQuant on SANA-1.6B, FLUX.1-schnell, and FLUX.1-dev with an $8\times$ smaller rank and up to $6.25\times$ faster quantization. Furthermore, on the Qwen3-8B LLM, rank-4 *H-SVDQuant* improves MMLU accuracy from 61.50% to 68.17% over rank-32 SVDQuant. Overall, *H-SVDQuant* achieves better W4A4 performance with substantially lower rank and quantization cost.

1 INTRODUCTION

Diffusion transformers (DiTs) (Peebles & Xie, 2023) repeatedly apply large linear transformations along the denoising trajectory, making low-precision inference an attractive route to reducing memory traffic and arithmetic cost. Weight–activation 4-bit post-training quantization (W4A4 PTQ) reduces both weights and activations to low precision without retraining, but the resulting errors can accumulate across layers and amplify over denoising steps.

SVDQuant (Li et al., 2025) first extracts a truncated-SVD branch and then quantizes the remaining residual. LRC (Scetbon & Hensman, 2024) jointly optimizes a quantized weights and low-rank correction. LoRaQ (Bouquet et al., 2026) directly optimizes a low-rank perturbation to compensate for weight quantization error. GPTQ-intrinsic LoRA (Zhang & Saab, 2026) also incorporates low-rank correction into second-order quantization. Despite these advances, existing methods do not explicitly formulate a multivariable coupled optimization problem, nor derive an efficient algorithm that jointly captures weight- and activation-side quantization under a small rank budget. This leads to our research question: *how can we jointly model the variables governing low-rank-assisted W4A4 PTQ and solve the resulting problem efficiently to maximize the effectiveness of low-rank branches?*

We present *H-SVDQuant*, a unified framework for low-rank W4A4 PTQ that formulates a joint reconstruction objective, derives optimization-based solvers, and instantiates them in a practical calibration algorithm. In Figure 1(a), a smoothed weight matrix $\tilde{W} = L_1 L_2 + R$, where L_1, L_2 denote the low-rank component and $\tilde{R} = Q(R)$ is the quantized residual. L_1 specifies the protected input subspace with calibration-weighted initialization. Analytically eliminating L_2 yields

$$H_{\perp} = H - H L_1 (L_1^{\top} H L_1)^{\dagger} L_1^{\top} H, \quad H_{\perp} L_1 = 0, \quad (1)$$

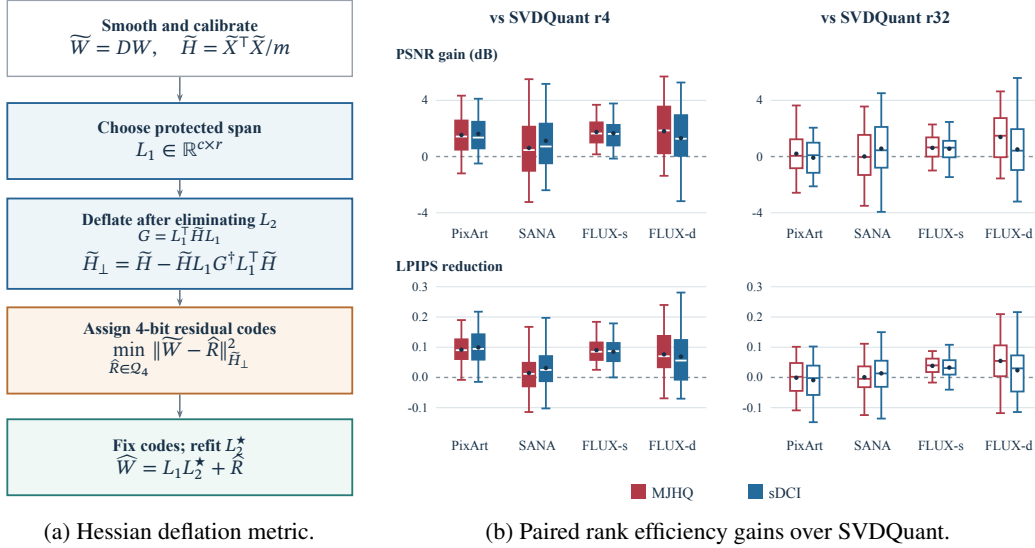


Figure 1: **(a) Residual quantization with Hessian deflation:** For a fixed protected subspace L_1 , Hessian deflation ($\tilde{H}_\perp L_1 = 0$) focuses residual quantization strictly on errors outside the low-rank subspace. **(b) Paired rank efficiency:** The paired distributions show consistent gains over rank-4 SVDQuant, and show model-dependent comparisons with rank-32 SVDQuant, with the strongest rank-efficiency gains on SANA and the FLUX variants.

where H is the calibration activation Gram matrix and $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse. We refer to this reduced geometry as *Hessian deflation*: Since $H_\perp L_1 = 0$, errors that lie in the protected input subspace can be compensated by the low-rank component. $\tilde{R}$ therefore only needs to account for the remaining errors measured by $H_\perp$. Then L_2 is refitted using a closed-form solution L_2^* . We optimize activation quantization through the diagonal smoothing and the activation-quantization surrogate. Specifically, we incorporate a smoothing matrix D and derive its ℓ_p -relaxed initialization with geometric programming, leaving the smoothed weights $\tilde{W} = DW$ for subsequent quantization, avoiding costly grid searches or online rotation overhead while accelerating calibration. Furthermore, our algorithm accommodates an activation-noise surrogate λ_A to suppress the amplification of activation quantization error during residual quantization, specified in Section 3.

We evaluate *H-SVDQuant* against PTQ baselines for DiTs: SVDQuant (Li et al., 2025), DiRotQ (Sharify et al., 2026), OrbitQuant (Lee et al., 2026), and ViDiT-Q (Zhao et al., 2025). Our backbones include diffusion transformers ranging from 0.6B to 20B parameters: PixArt- Σ (Chen et al., 2024), SANA-1.6B (Xie et al., 2025), FLUX.1-schnell (Black Forest Labs, 2024), FLUX.1-dev (Black Forest Labs, 2024), and Qwen-Image (Wu et al., 2025) (including its variant with 4-step Lightning LoRA). Under W4A4, *H-SVDQuant* consistently improves reconstruction fidelity across all backbones (up to +1.61 dB PSNR on PixArt- Σ) and rivals or surpasses the $8\times$ -larger rank-32 SVDQuant baseline on SANA-1.6B and FLUX models (Figure 1(b)). Evaluation on Qwen3-8B (Yang et al., 2025) also confirms that rank-4 *H-SVDQuant* achieves 10.44 WikiText-2 perplexity (PPL) and 68.17% MMLU accuracy (outperforming rank-32 SVDQuant at 11.21 PPL and 61.50% MMLU), suggesting that the proposed formulation also transfers to large language transformers (LLMs).

Our contributions are threefold:

- **Unified low-rank-assisted W4A4 PTQ problem.** We formulate a joint calibration objective for low-rank-assisted W4A4 PTQ that accounts for both weight- and activation-side quantization, seamlessly integrating an activation quantization surrogate objective.
- **Efficient optimization-based solvers.** We derive the optimization-based solvers for our framework, integrating calibration-weighted low-rank initialization, diagonal smoothing, second-order residual quantization, and closed-form low-rank refitting.
- **Extensive empirical validation under small-rank budgets.** We evaluate rank-4 *H-SVDQuant* across five diffusion backbones and Qwen3-8B, surpassing rank-32 SVDQuant on SANA and the FLUX variants while accelerating offline PTQ calibration by up to $6.25\times$.

2 RELATED WORK

Quantization of Diffusion Models. Early post-training quantization methods for diffusion models focused on the temporal non-stationarity induced by iterative denoising. Q-Diffusion addresses timestep-dependent activation distributions through timestep-aware calibration and split shortcut quantization, while PTQD explicitly models and corrects the accumulation of quantization noise along the denoising trajectory (Li et al., 2023; He et al., 2023). As transformer-based backbones became increasingly prominent in diffusion models, subsequent work identified additional challenges arising from activation outliers, timestep-dependent distributions, and heterogeneous layer sensitivity. ViDiT-Q (Zhao et al., 2025) introduces DiT-specific quantization schemes together with sensitivity-aware mixed-precision allocation. DiRotQ (Sharify et al., 2026) employs rotation-aware activation quantization to separate dominant activation directions from the low-bit subspace, whereas OrbitQuant (Lee et al., 2026) quantizes weights and activations in a normalized rotated representation using a shared non-uniform codebook.

Low-Rank-Assisted Quantization. Low-rank correction provides a complementary strategy for aggressive quantization by reserving a small auxiliary branch for structured error that is difficult to represent with low-bit codes. SVDQuant establishes this paradigm for diffusion models by concentrating weight outliers into a high-precision SVD branch and quantizing the remaining residual weights and activations to 4 bits (Li et al., 2025). Recent low-rank PTQ methods optimize different parts of this calibration problem. In language models, LRC jointly optimizes quantized weights with full-precision low-rank corrections acting on unquantized activations (Scetbon & Hensman, 2024). LoRaQ directly optimizes a low-rank approximation to quantization error, allowing the auxiliary branch to operate below 16-bit precision (Bouquet et al., 2026). GPTQ-intrinsic LoRA incorporates low-rank correction into an augmented-Hessian quantization pass (Zhang & Saab, 2026).

Activation-aware calibration PTQ. A separate line of work exploits calibration statistics to reshape the quantization problem. GPTQ uses approximate second-order information to minimize weight-quantization error under input-dependent curvature (Frantar et al., 2023), while AWQ identifies salient weights using activation statistics and protects them through channel-wise scaling (Lin et al., 2024). SmoothQuant migrates activation outliers into weights through an equivalent diagonal rescaling (Xiao et al., 2023). Rotation-based methods such as QuaRot and SpinQuant instead exploit orthogonal invariances to suppress activation outliers before low-bit quantization (Ashkboos et al., 2024; Liu et al., 2025).

Our work shares the use of low-rank compensation and activation-aware second-order information, but focuses on their coupled W4A4 formulation and studies this coupling under small auxiliary rank budgets for efficient PTQ in diffusion transformers.

3 METHOD

3.1 PROBLEM MODELING

For a linear layer $Y = XW$ with calibration input $X \in \mathbb{R}^{m \times c}$ and weight matrix $W \in \mathbb{R}^{c \times n}$, $r \ll \min(c, n)$ denotes the protected rank budget. We apply an equivalent diagonal reparameterization

$$\tilde{X} = XD^{-1}, \quad \tilde{W} = DW, \quad (2)$$

where $D = \text{diag}(d)$ with $d_i > 0$ is the diagonal smoothing matrix. This transform redistributes channel ranges between activations and weights while preserving $XW = \tilde{X}\tilde{W}$, where $\tilde{X}, \tilde{W}$ are the smoothed activations and weights.

Low-rank-assisted operator. We adopt a low-rank-assisted operator with a rank- r high-precision branch, keeping $\tilde{X}L_1L_2$ separate from the low-bit path. The resulting operator is

$$\hat{Y} = \tilde{X}L_1L_2 + Q_A(\tilde{X})\hat{R}, \quad (3)$$

where $L_1 \in \mathbb{R}^{c \times r}$ and $L_2 \in \mathbb{R}^{r \times n}$ are the input- and output-side low-rank factors, respectively, and $\hat{R}$ denotes the quantized residual. We then derive the reconstruction objective given by this low-rank-assisted operator.

Reconstruction objective. Defining remaining weight error $C = \widetilde{W} - L_1 L_2 - \widehat{R}$ and activation error $E_A = Q_A(\widetilde{X}) - \widetilde{X}$, Equation (3) gives exact layer-output error

$$Y - \widehat{Y} = \widetilde{X}C - E_A \widehat{R}. \quad (4)$$

Let $\widetilde{H} = \frac{1}{m} \widetilde{X}^\top \widetilde{X}$ and $\Sigma_A = \frac{1}{m} E_A^\top E_A$, with $\|B\|_M^2 \triangleq \text{tr}(B^\top M B)$ for PSD matrix M . Expanding Equation (4) yields

$$\frac{1}{m} \|Y - \widehat{Y}\|_F^2 = \|C\|_{\widetilde{H}}^2 + \|\widehat{R}\|_{\Sigma_A}^2 - \frac{2}{m} \text{tr} \left(C^\top \widetilde{X}^\top E_A \widehat{R} \right). \quad (5)$$

Under standard conditional zero-mean uniform rounding assumptions, the cross term vanishes in expectation ($\mathbb{E}[\text{tr}(C^\top \widetilde{X}^\top E_A \widehat{R})] = 0$), yielding $\frac{1}{m} \|Y - \widehat{Y}\|_F^2 \approx \|C\|_{\widetilde{H}}^2 + \|\widehat{R}\|_{\Sigma_A}^2$. For a b_A -bit symmetric quantizer with $\kappa_A = 2^{b_A-1} - 1$ and activation group $\mathcal{G}(i)$ for channel i , we model activation-noise covariance via the block-constant diagonal surrogate:

$$\overline{\Sigma}_A = \text{Diag}(s_A), \quad (s_A)_i = \frac{1}{12\kappa_A^2} \mathbb{E}_t \left[\max_{j \in \mathcal{G}(i)} \widetilde{X}_{tj}^2 \right]. \quad (6)$$

Introducing weighting parameter $\lambda_A \geq 0$, the joint calibration objective is

$$\mathcal{L}(L_1, L_2, \widehat{R}) = \frac{1}{2} \left\| \widetilde{W} - L_1 L_2 - \widehat{R} \right\|_{\widetilde{H}}^2 + \frac{1}{2} \lambda_A \left\| \widehat{R} \right\|_{\overline{\Sigma}_A}^2. \quad (7)$$

Equation (7) unifies low-rank compensation and activation-noise suppression under a shared calibration framework where the first term minimizes weight error, while the second suppresses activation-error amplification through $\widehat{R}$. We use Equation (7) as the common objective underlying the derivations in Section 3.2; $\lambda_A = 0$ gives its weight-centric specialization, while $\lambda_A > 0$ enables activation-aware code optimization when activation-quantization error is propagated during calibration.

3.2 H-SVDQUANT

We now derive tractable updates for the coupled objective in Equation (7). The derivation proceeds by analytically eliminating the refittable factor L_2 , reducing residual-code optimization to a second-order metric-and-target problem, while deriving calibration-aware updates for the protected subspace and smoothing transform.

Hessian deflation. For fixed D , L_1 , and $\widehat{R}$, minimizing over L_2 yields the closed-form weighted least-squares solution:

$$L_2^* = \left(L_1^\top \widetilde{H} L_1 \right)^\dagger L_1^\top \widetilde{H} \left(\widetilde{W} - \widehat{R} \right), \quad (8)$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose pseudoinverse.

Proposition 1 (Hessian deflation). Substituting L_2^* into Equation (7) replaces $\widetilde{H}$ with the deflated Hessian

$$\widetilde{H}_\perp = \widetilde{H} - \widetilde{H} L_1 \left(L_1^\top \widetilde{H} L_1 \right)^\dagger L_1^\top \widetilde{H}, \quad \text{satisfying} \quad \widetilde{H}_\perp L_1 = 0. \quad (9)$$

The derivation of L_2^* and proof are provided in Appendix B. The identity $\widetilde{H}_\perp L_1 = 0$ shows that errors compensable by the protected low-rank component are eliminated from the residual objective; residual quantization thus targets only the remaining reconstruction error.

Residual-code optimization. After eliminating L_2 , the code-dependent objective becomes

$$\mathcal{L}_{\text{code}} = \frac{1}{2} \left\| \widetilde{W} - \widehat{R} \right\|_{\widetilde{H}_\perp}^2 + \frac{1}{2} \lambda_A \left\| \widehat{R} \right\|_{\overline{\Sigma}_A}^2 = \frac{1}{2} \left\| \widehat{R} - T_Q \right\|_{M_Q}^2 + \text{const}, \quad (10)$$

where

$$M_Q = \widetilde{H}_\perp + \lambda_A \overline{\Sigma}_A, \quad T_Q = M_Q^\dagger \widetilde{H}_\perp \widetilde{W}. \quad (11)$$

We use Equation (6) as the activation-noise surrogate: M_Q reshapes the metric geometry, while T_Q adjusts the effective approximation target.

Low-rank initialization. Rather than initializing the protected branch with ordinary truncated SVD, we solve a calibration-weighted rank- r approximation $\min_{\text{rank}(L) \leq r} \frac{1}{2} \|\tilde{H}_{\text{reg}}^{1/2}(\tilde{W} - L)\|_F^2$, whose exact solution is , where $\tilde{H}_{\text{reg}} = \tilde{H} + \tau I$ with $\tau > 0$ is a positive-definite regularization of the calibration Gram matrix. The exact solution to this regularized approximation problem is

$$L^* = \tilde{H}_{\text{reg}}^{-1/2} \left[\tilde{H}_{\text{reg}}^{1/2} \tilde{W} \right]_r, \quad (12)$$

where $[\cdot]_r$ denotes rank- r truncated SVD. We initialize L_1 with the resulting input-side span under an $\tilde{H}_{\text{reg}}$ -orthonormal gauge, preserving $L_1 L_2$ while improving numerical conditioning. Details are in Appendix C.

Diagonal smoothing. The diagonal reparameterization in Equation (2) redistributes channel ranges without changing the full-precision linear map. Given residual weight P and diagonal curvature h_i , a closed-form ℓ_p relaxation of the grouped geometric-program objective yields

$$d_i \propto \left(h_i / \sum_j |P_{ij}|^p \right)^{1/(p+2)}. \quad (13)$$

We normalize the resulting scales by their geometric mean and optionally refine them in log space using the grouped objective; Appendix D gives the exact GP formulation and the derivation of the closed-form relaxation.

3.3 IMPLEMENTED ALGORITHM

Based on the core derivations and formulas, we compose the conditional updates into the overall calibration algorithm and describe its numerical treatment and deployed operator.

Algorithm workflow. As summarized in Appendix Algorithm 1, the unsmoothed calibration matrix $H_0 = X^\top X/m$ is accumulated once. At each outer iteration, we update smoothing matrix D , obtain $\tilde{H} = D^{-1} H_0 D^{-1}$, initialize L_1 , construct the deflated Hessian $\tilde{H}_\perp$, quantize residual $\tilde{R}$, and refit L_2 in closed form. With cached activations, candidate states are compared using empirical W4A4 output MSE with the dynamic quantizer. For activation-aware W4A4 calibration, we use a GPTQ-style solver with M_Q and T_Q replacing the conventional curvature matrix and weight target. When $\lambda_A = 0$, the solver reduces to weight-only residual quantization under $\tilde{H}_\perp$, followed by the same closed-form refit of L_2 . The two settings optimize the same underlying low-rank-assisted operator but differ in whether activation-noise amplification is included in residual-code selection.

Numerical stability and damping. The identities in Eqs. (9)–(10) describe the undamped formulation. Since $\tilde{H}_\perp$ and M_Q can be singular, practical solves use small diagonal damping ($M_Q + \epsilon I$) before Cholesky factorization or inversion when computing T_Q , which slightly perturbs $\tilde{H}_\perp L_1 = 0$ while ensuring stability.

4 EXPERIMENTAL SETUP

Models. We benchmark *H-SVDQuant* across: PixArt- Σ (0.6B) (Chen et al., 2024), SANA-1.6B (Xie et al., 2025), FLUX.1-dev (12B) (Black Forest Labs, 2024), and the Qwen-Image (20B) (Wu et al., 2025) (50-step base model). Accelerated few-step variants (the 4-step FLUX.1-schnell and 4-step Qwen-Image Lightning) are evaluated in Appendix Table 8, and visual generation comparisons for both base and Lightning models are detailed in Appendix G. Additionally, we evaluate Qwen3-8B (Yang et al., 2025) as a controlled testbed for rank efficiency in large language models.

Implementation details. All diffusion backbones operate in the W4A4 post-training quantization regime with group size 64 and an auxiliary protected rank $r = 4$ (FP16/BF16). Activations are quantized using per-token, per-group 4-bit quantization. In our primary DiT experiments, we use reference activations across transformer blocks with $\lambda_A = 0$ (except for PixArt- Σ at $\lambda_A = 0.05$ due to its different activation geometry) in Table 1. We sequentially replay activations within each block as module groups are quantized and replaced. These can be found in our Algorithm 1. We analyze the effect and strength of λ_A in Section 6, and compare different strategies for propagating quantized activations and accumulating quantization error in Appendix I.

Datasets. Following SVDQuant (Li et al., 2025), we evaluate the generation fidelity and quality of quantized diffusion models on the category-balanced **MJHQ-30K** (Li et al., 2024) and the diverse **sDCI** (Urbanek et al., 2024) benchmarks. For each diffusion backbone, evaluation is performed on fixed **MJHQ-5K** and **sDCI-1K** subsets. Each prompt is paired with the output of the corresponding native high-precision teacher model using identical random seeds, initial latents, image resolution (1024×1024), and scheduler steps. For diffusion-model calibration, we use the same set of 128 prompts released in the official SVDQuant repository. For Qwen3-8B calibration, we sample 128 sequences of 512 tokens each from the WikiText-2 training split. We report WikiText-2 (Merity et al., 2017) perplexity together with zero-shot accuracy on five standard benchmarks: MMLU (Hendrycks et al., 2021), ARC-Easy and ARC-Challenge (Clark et al., 2018), HellaSwag (Zellers et al., 2019), and PIQA (Bisk et al., 2020).

Baselines. We compare *H-SVDQuant* against representative post-training quantization (PTQ) baselines. **SVDQuant** (Li et al., 2025): we use rank-4 SVDQuant as the primary matched-capacity baseline and additionally include rank-32 SVDQuant, which uses an $8\times$ larger protected-rank budget, as a higher-capacity reference. **DiRotQ** (Sharify et al., 2026): we evaluate its W4A4 variant while matching the fraction of high-precision values to that of our method. **OrbitQuant** (Lee et al., 2026) and **ViDiT-Q** (Zhao et al., 2025): we adapt both methods to the same W4A4 setting with group size 64 and per-token activation quantization for a controlled comparison.

Metrics. Following previous works (Li et al., 2025), we assess generation fidelity and quality using four standard metrics: **Peak Signal-to-Noise Ratio (PSNR, $\uparrow$)** measures pixel-level numerical reconstruction fidelity with respect to the 16-bit teacher model; **Learned Perceptual Image Patch Similarity (LPIPS, $\downarrow$)** (Zhang et al., 2018)) evaluates perceptual similarity against prompt-paired teacher outputs; **CLIP Score (CLIP, $\uparrow$)** (Hessel et al., 2021) measures text–image semantic alignment with the input prompts; and **ImageReward (IR, $\uparrow$)** (Xu et al., 2023) approximates human visual preference for generated images. For language modeling on Qwen3-8B, we report WikiText-2 **Perplexity (PPL, $\downarrow$)** and zero-shot accuracy. For the PixArt- Σ ablation in Table 5a, we additionally report Structural Similarity (SSIM, $\uparrow$) (Wang et al., 2004).

5 MAIN RESULTS ON DIFFUSION TRANSFORMERS

We report quantitative W4A4 results in Table 1 across DiT backbones, and show corresponding qualitative visual comparisons in Figure 2. Note that PSNR and LPIPS measure teacher fidelity, while CLIP and IR measure standalone generation quality. They measure different properties; thus the method with the best PSNR need not have the highest CLIP or ImageReward score.

Quantitative Comparison. As shown in Table 1, under W4A4 *H-SVDQuant* consistently improves teacher fidelity over SVDQuant on both MJHQ and sDCI, achieving higher PSNR and lower LPIPS on every evaluated backbone. On PixArt- Σ , *H-SVDQuant* improves over rank-4 SVDQuant by +1.53 dB PSNR and -0.092 LPIPS on MJHQ, and by +1.61 dB and -0.099 LPIPS on sDCI. Its teacher-fidelity results are close to rank-32 SVDQuant and DiRotQ, while CLIP and ImageReward vary across methods. On SANA-1.6B, *H-SVDQuant* reaches 20.25/0.161 PSNR/LPIPS on MJHQ and 18.25/0.185 on sDCI, improving substantially over rank-4 SVDQuant and matching or exceeding rank-32 SVDQuant in teacher-fidelity metrics. On FLUX.1-dev, *H-SVDQuant* improves rank-4 SVDQuant by +1.81 dB PSNR and -0.076 LPIPS on MJHQ, and by +1.32 dB and -0.069 LPIPS on sDCI. It also exceeds rank-32 SVDQuant in PSNR and LPIPS on both subsets. On Qwen-Image, *H-SVDQuant* improves teacher fidelity over rank-4 SVDQuant on both subsets. Relative to rank-32 SVDQuant, the comparison is mixed on MJHQ—*H-SVDQuant* obtains lower LPIPS but slightly lower PSNR—while it improves both PSNR and LPIPS on sDCI. Extended quantitative comparisons on accelerated variants are provided in Appendix Table 8.

Qualitative Comparison. Figure 2 provides visual comparisons across representative prompts under identical random seeds. OrbitQuant and ViDiT-Q struggle under 4-bit activations, exhibiting pervasive noise, structural distortion, and contrast loss (e.g., distorted geometry on PixArt- Σ and noisy surfaces on SANA-1.6B and FLUX.1-dev). While DiRotQ and rank-32 SVDQuant achieve better visual quality, they lose fidelity in a few cases (e.g., different elephant direction and hill space, over-detailed lion). In contrast, *H-SVDQuant* consistently preserves sharp object boundaries and

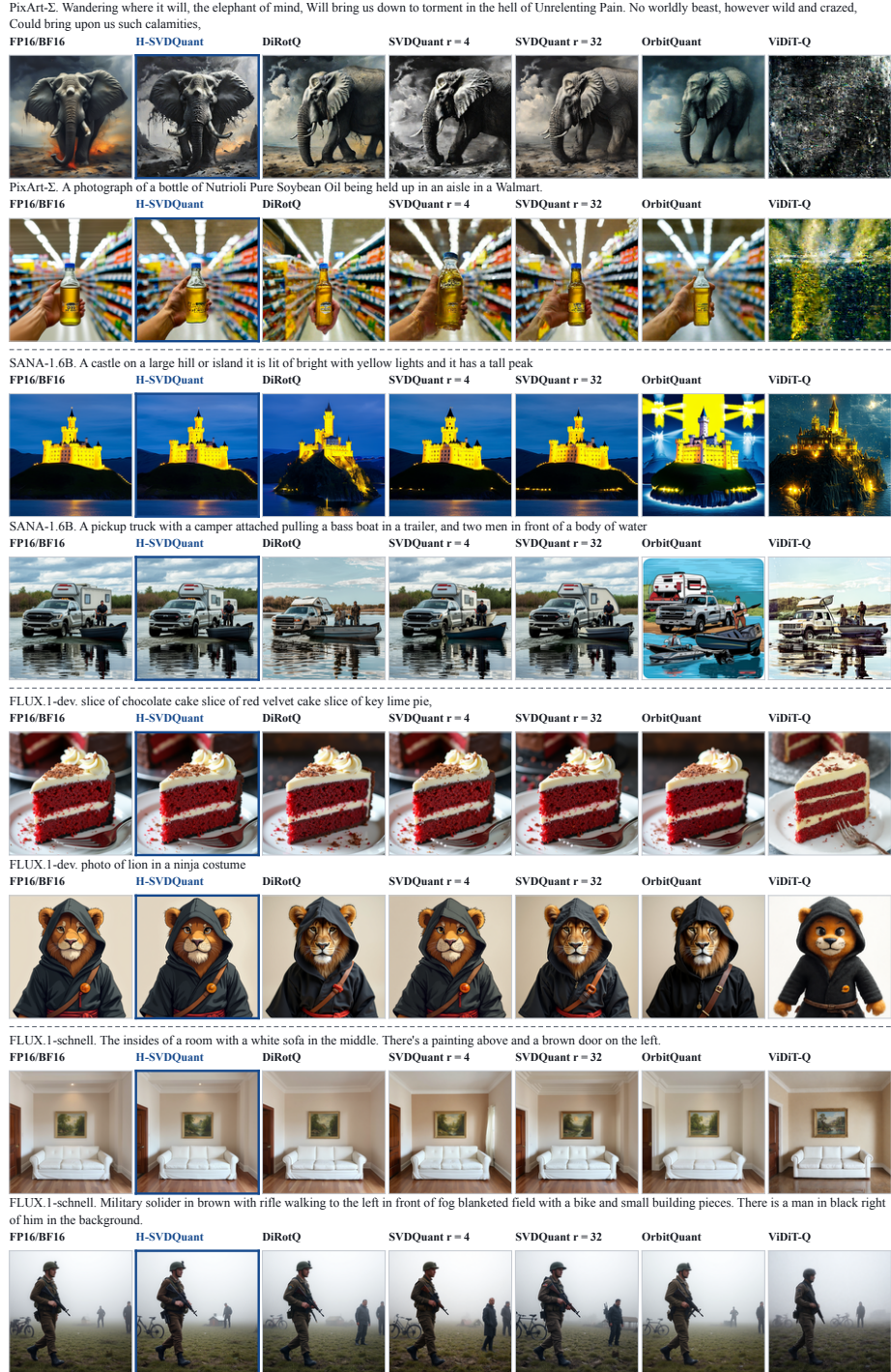


Figure 2: **W4A4 reconstructions across diffusion transformers.** Each row compares dense FP16/BF16 against quantized methods under identical prompt and seed. Blue boxes highlight *H-SVDQuant* ($r = 4$). Across all backbones, *H-SVDQuant* preserves textures, fine structures, and prompt alignment faithful to the 16-bit reference.

Table 1: **W4A4 fidelity and generation quality across modern diffusion transformer backbones.** Bold and underline denote the best and second-best results, respectively. All evaluated methods use 4-bit weights and 4-bit activations (W4A4) with group-64.

Method	MJHQ				sDCI			
	PSNR $\uparrow$	LPIPS $\downarrow$	CLIP $\uparrow$	IR $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	CLIP $\uparrow$	IR $\uparrow$
PixArt-Σ								
H-SVDQuant $r = 4$	16.47	<u>0.386</u>	<u>25.995</u>	0.8852	15.36	0.422	26.203	<u>0.9816</u>
SVDQuant $r = 4$	14.94	0.478	25.832	0.7489	13.75	0.521	25.604	0.8514
SVDQuant $r = 32$	16.27	0.385	25.962	0.8950	15.45	0.413	25.718	0.9002
DiRotQ	<u>16.36</u>	0.389	26.047	0.9165	<u>15.39</u>	<u>0.416</u>	<u>26.239</u>	1.0417
OrbitQuant	14.88	0.488	25.703	0.7711	14.00	0.528	26.441	0.8695
SANA-1.6B								
H-SVDQuant $r = 4$	20.25	0.161	25.996	1.1053	18.25	0.185	26.259	1.0724
SVDQuant $r = 4$	19.64	0.176	25.956	1.1213	17.13	0.216	<u>26.336</u>	1.0750
SVDQuant $r = 32$	<u>20.24</u>	<u>0.162</u>	25.931	1.0994	<u>17.69</u>	<u>0.199</u>	26.330	1.1102
DiRotQ	19.83	0.173	<u>26.051</u>	<u>1.1064</u>	17.61	0.209	26.274	<u>1.1059</u>
OrbitQuant	14.48	0.415	26.124	1.0465	12.10	0.495	26.701	1.0809
ViDiT-Q	13.38	0.444	25.721	0.9871	11.44	0.483	25.783	0.8971
FLUX.1-dev								
H-SVDQuant $r = 4$	<u>20.58</u>	<u>0.227</u>	23.078	0.9917	<u>18.93</u>	<u>0.250</u>	24.586	1.0205
SVDQuant $r = 4$	18.77	0.303	<u>23.156</u>	0.9771	17.61	0.319	<u>24.675</u>	1.0630
SVDQuant $r = 32$	19.19	0.282	23.107	0.9782	18.42	0.274	24.730	<u>1.0594</u>
DiRotQ	20.95	0.224	23.534	0.9864	19.92	0.218	24.471	1.0181
OrbitQuant	19.44	0.279	22.877	0.9637	18.25	0.284	24.496	1.0476
ViDiT-Q	15.68	0.518	22.122	0.6232	14.49	0.539	24.140	0.7236
Qwen-Image (50 steps)								
H-SVDQuant $r = 4$	19.23	<u>0.236</u>	29.669	1.5775	18.44	0.254	25.817	1.2027
SVDQuant $r = 4$	18.78	0.256	29.714	1.5540	17.70	0.279	<u>25.755</u>	1.2191
SVDQuant $r = 32$	<u>19.37</u>	0.243	<u>29.734</u>	<u>1.5665</u>	18.03	0.272	25.681	<u>1.2095</u>
DiRotQ	19.63	0.218	29.769	1.5404	<u>18.39</u>	<u>0.257</u>	25.642	1.2080

Table 2: **Recorded rank-4 PTQ cost on NVIDIA H20.** Time is each checkpoint’s complete launcher wall interval; peak memory is the recorded CUDA allocation.

Model	Method	Minutes	Speedup vs. SVDQuant $r=4$	Peak GiB
SANA-1.6B	H-SVDQuant	4.8	$6.25\times$	9.9
	SVDQuant	30.0	–	11.7
FLUX-schnell	H-SVDQuant	282.4	$1.81\times$	52.0
	SVDQuant	511.1	–	50.7
FLUX-dev	H-SVDQuant	285.6	$1.85\times$	52.0
	SVDQuant	527.4	–	51.3

subtle textures faithful to the 16-bit reference teacher. Extended visual comparisons on both base and Lightning Qwen-Image models are provided in Appendix G.

Rank Efficiency and Calibration Cost. The rank advantage is clearest on SANA and the FLUX backbones. Besides the mean values, we also provide the paired distributions in Figure 1(b) and Appendix Figure 5 for a clearer comparison. As shown in Table 2, *H-SVDQuant* reduces offline PTQ calibration cost in the recorded base runs. On SANA-1.6B, the measured wall time decreases from 30.0 to 4.8 minutes, while the FLUX.1-schnell and FLUX.1-dev base runs show $1.81\times$ and $1.85\times$ speedups, respectively.

6 ANALYSIS AND ABLATION STUDIES

In this section, we conduct in-depth diagnostics and ablation studies using PixArt- Σ (for vision generation) and Qwen3-8B (for language modeling). Unless stated, our optional cached iteration of D is 20 and the outer iteration is set to be 2. In Qwen analysis, we disable the optional cached optimization and set outer iteration as 1 in Table 4(a), 5(b), 6 and set outer iteration as 2 in Table 7.

Table 3: **Qwen3-8B zero-shot evaluation (W4A4, group-128)**. Mean denotes the unweighted average across all five downstream tasks. SVDQuant refers to the local language-model adaptation; FP16 serves as dense reference.

Method	MMLU	ARC-C	ARC-E	HellaSwag	PIQA	Mean
FP16	72.95	56.40	83.54	74.97	76.88	72.95
SVDQuant adaptation $r = 4$	61.34	45.31	73.02	67.83	71.76	63.85
SVDQuant adaptation $r = 32$	61.50	48.89	74.83	67.84	72.20	65.05
<i>H-SVDQuant</i> $r = 4$	68.17	49.83	78.54	70.30	73.34	68.04

Table 4: **Rank efficiency and activation-error propagation on Qwen3-8B** (W4A4, group-128, $\lambda_A = 0.25$). (a) WikiText-2 perplexity across different protected ranks. (b) Relative activation MSE averaged over 36 transformer blocks using eight 512-token sequences (measured against 16-bit reference activations). Local MSE measures per-block quantization error under reference inputs, while propagated MSE additionally captures errors accumulated from preceding quantized blocks.

(a) Perplexity across protected ranks

Method	Rank	PPL ↓
SVDQuant adapt.	4	11.93
SVDQuant adapt.	32	11.21
<i>H-SVDQuant</i>	0	10.72
<i>H-SVDQuant</i>	2	10.48
<i>H-SVDQuant</i>	4	10.44

(b) Local vs. propagated activation error

Method	Local MSE ↓	Prop. MSE ↓
SVDQuant adapt. $r = 4$	0.00215	0.02068
SVDQuant adapt. $r = 32$	0.00199	0.01715
<i>H-SVDQuant</i> $r = 4$	0.00150	0.01194
Reduction vs. $r = 4$	−30.2%	−42.3%
Reduction vs. $r = 32$	−24.6%	−30.4%

We explore the effect of rank r and λ_A , and demonstrate the advantages of our three optimization components: $\mathbf{H}_\perp$, $\mathbf{L}_1$, $\mathbf{D}$.

Rank Sweep Ablation. We first examine the protected low-rank budget r . As detailed in Table 4(a), rank-0 *H-SVDQuant* already achieves 10.72 perplexity, substantially outperforming rank-32 SVDQuant (11.21). Introducing a small protected branch provides an additional improvement, reducing perplexity to 10.44 (10.438). As shown in Table 4(b), the rank-4 configuration also exhibits substantially lower local and propagated activation error than the SVDQuant baselines. Together, these results show that *H-SVDQuant* remains effective under very small protected-rank budgets, characterizing the complete *H-SVDQuant* recipe rather than attributing the gain to a single component.

Activation-Noise Sensitivity. We ablate the activation-noise weighting parameter in Equation (7) across both PixArt- Σ and Qwen3-8B (Table 5), with $\lambda_A \in \{0, 0.05, 0.10, 0.25, 0.50\}$. On PixArt- Σ (Table 5(a)), λ_A has a modest effect. A small activation-noise weight improves all three metrics at once: they attain their best values at $\lambda_A = 0.05$ (15.99 dB PSNR, 0.583 SSIM, and 0.414 LPIPS), a +0.29 dB PSNR gain over $\lambda_A = 0$. Raising λ_A further gives this gain back, and $\lambda_A = 0.50$ falls slightly below the unweighted baseline (15.37 dB PSNR, 0.554 SSIM, 0.472 LPIPS). On Qwen3-8B (Table 5(b)), WikiText-2 perplexity remains highly stable across small weighting coefficients ($\lambda_A \leq 0.25$), reaching its lowest value of 10.433 at $\lambda_A = 0.05$ and ranking second at $\lambda_A = 0.25$ (10.438). Empirical results demonstrate that moderate activation-aware weighting ($\lambda_A \in [0.05, 0.25]$) provides the most effective balance, whereas excessive weighting ($\lambda_A = 0.50$) leads to performance regression. The benefit of activation-aware weighting is therefore regime- and model-dependent.

Hessian Deflation and Protected-Subspace Initialization. Table 6 presents a controlled 2×2 ablation on Qwen3-8B ($\lambda_A = 0$). We fix $\mathbf{D}$, r , the quantizer, GPTQ settings, calibration data, and $\mathbf{L}_2$ refitting, varying only the $\mathbf{L}_1$ initialization (plain SVD vs. calibration-weighted, Equation (12)) and the GPTQ metric ($\widetilde{\mathbf{H}}$ vs. $\widetilde{\mathbf{H}}_\perp$, Equation (9)). For each fixed $\mathbf{L}_1$, both metrics share the same initial residual. Holding $\mathbf{L}_1$ fixed, Hessian deflation reduces PPL by 0.020 (plain SVD) and 0.006 (weighted SVD). Holding the GPTQ metric fixed, weighted initialization reduces PPL by 0.067 ($\widetilde{\mathbf{H}}$) and 0.053 ($\widetilde{\mathbf{H}}_\perp$). Their combination achieves the lowest PPL (10.486, reducing baseline PPL by 0.073), supporting their complementary roles: weighted initialization selects protected directions, while deflation excludes them from residual-code optimization.

Table 5: **Activation-noise-weight sensitivity on PixArt- Σ (group-64), and Qwen3-8B (group-128). (W4A4, $r = 4$).** We evaluating sensitivity to activation-noise weighting parameter λ_A across vision generation (PixArt- Σ) and language modeling (Qwen3-8B). Bold denotes best value in each backbone. In Qwen, we disable the optional cached iteration and the outer iteration, the result of $\lambda_A = 0$ matches Table 6.

(a) PixArt- Σ				(b) Qwen3-8B	
λ_A	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	λ_A	WikiText-2 PPL $\downarrow$
0	15.70	0.557	0.467	0	10.486
0.05	15.99	0.583	0.414	0.05	10.433
0.10	15.97	0.574	0.441	0.10	10.454
0.25	15.73	0.580	0.433	0.25	<u>10.438</u>
0.50	15.37	0.554	0.472	0.50	10.640

Table 6: **Controlled ablation of protected-subspace initialization (in Equation (12).) and Hessian deflation on Qwen3-8B (W4A4, group-128, $\lambda_A = 0$).** We keep all other calibration settings fixed. Each row of $\widetilde{H}/\widetilde{H}_\perp$ runs shares the same initialized L_1 and residual. Lower WikiText-2 PPL is better.

L_1 initialization	Original $\widetilde{H}$	Deflated $\widetilde{H}_\perp$
Plain SVD	10.559	10.539
Calibration-weighted	10.492	10.486

Table 7: **Effect of our closed-form smoothing solution on Qwen3-8B (W4A4/W4A16, group-128, $\lambda_A = 0.25$).** We keep the H -SVDQuant pipeline fixed and replace the SVDQuant grid-searched D with our closed-form solution in Equation (13). We disable the optional cached optimization of D and only use its initialization. Avg. denotes the mean accuracy over MMLU, ARC-Challenge, ARC-Easy, HellaSwag, and PIQA. Lower PPL and higher accuracies are better.

D construction	Quant.	PPL $\downarrow$	MMLU $\uparrow$	ARC-C $\uparrow$	ARC-E $\uparrow$	Hella. $\uparrow$	PIQA $\uparrow$	Avg. $\uparrow$
Grid-search D (SVDQuant)	W4A4	10.895	66.59	51.11	79.21	68.26	74.05	67.84
Closed-form D (Ours)	W4A4	10.469	67.23	50.77	78.16	70.77	75.24	68.44
Grid-search D (SVDQuant)	W4A16	10.078	70.83	53.75	82.03	73.80	76.82	71.45
Closed-form D (Ours)	W4A16	9.976	70.94	54.18	82.37	73.47	77.09	71.61

Closed-Form Diagonal Smoothing. We isolate the effect of our smoothing solver by keeping the H -SVDQuant pipeline fixed and changing only how D is obtained. Specifically, we compare (i) SVDQuant-style grid-searched D , and (ii) our analytical solution in Equation (13). As shown in Table 7, the closed-form solution improves the overall result under both quantization settings: for W4A4, it reduces PPL from 10.895 to 10.469 and improves the average accuracy from 67.84 to 68.44; for W4A16, it reduces PPL from 10.078 to 9.976 and improves the average accuracy from 71.45 to 71.61. The gains are particularly clear on HellaSwag and PIQA under W4A4, with improvements of +2.51 and +1.19 points, respectively, while W4A16 also improves ARC-Challenge and ARC-Easy by +0.43 and +0.34. This controlled comparison provides empirical support for Equation (13): the analytically derived D provides a better overall calibration point than SVDQuant-style grid search while eliminating the associated search overhead.

7 CONCLUSION

In this paper, we presented H -SVDQuant, a unified framework for low-rank-assisted W4A4 PTQ. By jointly modeling low-rank compensation, residual quantization, and activation noise, we derive a deflated Hessian through analytical elimination of the output-side low-rank factor, together with an activation-aware extension. Our efficient calibration algorithm combines weighted low-rank initialization, GP-derived smoothing, GPTQ-style quantization, and closed-form refitting. Across diffusion transformers, rank-4 H -SVDQuant consistently improves PSNR and LPIPS over rank-4 SVDQuant, matching or outperforming rank-32 SVDQuant on SANA-1.6B and FLUX variants with an $8\times$ smaller rank. It also reduces offline calibration time by up to $6.25\times$. Results on Qwen3-8B further demonstrate improved perplexity and downstream accuracy over rank-32 SVDQuant, suggesting broader applicability beyond diffusion transformers.

ACKNOWLEDGMENTS

The authors would like to thank Jian Yang from Renmin University of China for his insightful discussions and general help.

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Appendix

A PROBLEM REFORMULATION AND ERROR DECOMPOSITION

This section uses unnormalized Gram matrices and summed reconstruction losses; dividing both weight and activation terms by the calibration sample count recovers the main-text convention. The provisional-residual expansion below motivates the smoothing surrogate. Once codes are selected, the residual matrix is held fixed while L_2 is refitted, as in Eq. (8).

Consider a linear layer

$$Y = XW, \quad X \in \mathbb{R}^{m \times c}, \quad W \in \mathbb{R}^{c \times n}.$$

A.1 SMOOTHING AND LOW-RANK DECOMPOSITION

Let

$$D = \text{diag}(d_1, \dots, d_c), \quad d_i > 0,$$

and define

$$\tilde{X} = XD^{-1}, \quad \tilde{W} = DW.$$

Then

$$Y = XW = \tilde{X}\tilde{W}.$$

Define

$$H = X^\top X, \quad \tilde{H} = \tilde{X}^\top \tilde{X} = D^{-1}HD^{-1}.$$

We decompose the smoothed weight as

$$\boxed{\tilde{W} = L_1 L_2 + R,} \quad R = \tilde{W} - L_1 L_2,$$

where

$$L_1 \in \mathbb{R}^{c \times r}, \quad L_2 \in \mathbb{R}^{r \times n}.$$

The quantized forward pass is

$$\boxed{\hat{Y} = \tilde{X} L_1 L_2 + Q_A(\tilde{X}) Q_W(R).}$$

A.2 ERROR DECOMPOSITION

Define the activation and weight quantization errors as

$$E_A = Q_A(\tilde{X}) - \tilde{X}, \quad \Delta_W = R - Q_W(R).$$

Hence

$$Q_A(\tilde{X}) = \tilde{X} + E_A, \quad Q_W(R) = R - \Delta_W.$$

Substituting these expressions into the quantized forward pass gives

$$\begin{aligned} \hat{Y} &= \tilde{X} L_1 L_2 + (\tilde{X} + E_A)(R - \Delta_W) \\ &= Y - \tilde{X} \Delta_W + E_A R - E_A \Delta_W. \end{aligned} \tag{14}$$

Therefore,

$$Y - \hat{Y} = \tilde{X} \Delta_W - E_A R + E_A \Delta_W.$$

Neglecting the second-order term $E_A \Delta_W$,

$$\boxed{Y - \hat{Y} \approx \tilde{X} \Delta_W - E_A R.} \tag{15}$$

Under the additive-noise assumption, the cross term vanishes in expectation, yielding

$$\boxed{\mathcal{L} \approx F_W + F_A,} \tag{16}$$

where

$$\boxed{F_W = \text{tr} \left(\Delta_W^\top \tilde{H} \Delta_W \right),} \tag{17}$$

and

$$F_A = \text{tr} (R^\top \Sigma_A R), \quad \Sigma_A = \mathbb{E}[E_A^\top E_A]. \quad (18)$$

Thus,

$$\mathcal{L} \approx \text{tr} (\Delta_W^\top \tilde{H} \Delta_W) + \text{tr} (R^\top \Sigma_A R). \quad (19)$$

B ELIMINATION OF L_2 AND HESSIAN DEFLATION

The inverse formulas below assume a nonsingular protected Gram matrix. For the general positive-semidefinite case, replace the inverse with its Moore–Penrose pseudoinverse: the residual is the orthogonal projection onto the complement of $\text{range}(\tilde{H}^{1/2} L_1)$. Consequently the same minimized metric and annihilation identity hold.

Fix D , L_1 , and the quantized residual $Q_W(R)$. For output column j , define

$$\tilde{w}_j = \tilde{W}_{:,j}, \quad \hat{w}_j = Q_W(R)_{:,j}, \quad \ell_j = (L_2)_{:,j},$$

and let

$$u_j = \tilde{w}_j - \hat{w}_j.$$

The reconstruction problem associated with column j is

$$\min_{\ell_j \in \mathbb{R}^r} \frac{1}{2} \left\| \tilde{X} (u_j - L_1 \ell_j) \right\|_2^2. \quad (20)$$

Using

$$\tilde{H} = \tilde{X}^\top \tilde{X},$$

this is equivalently

$$\min_{\ell_j \in \mathbb{R}^r} \frac{1}{2} (u_j - L_1 \ell_j)^\top \tilde{H} (u_j - L_1 \ell_j).$$

Define

$$M = L_1^\top \tilde{H} L_1.$$

Expanding the objective gives

$$f(\ell_j) = \frac{1}{2} u_j^\top \tilde{H} u_j - \ell_j^\top L_1^\top \tilde{H} u_j + \frac{1}{2} \ell_j^\top M \ell_j.$$

The first-order optimality condition is

$$-L_1^\top \tilde{H} u_j + M \ell_j = 0,$$

hence

$$\ell_j^* = M^{-1} L_1^\top \tilde{H} u_j. \quad (21)$$

Substituting ℓ_j^* back into the objective yields

$$\begin{aligned} f(\ell_j^*) &= \frac{1}{2} u_j^\top \tilde{H} u_j - \frac{1}{2} u_j^\top \tilde{H} L_1 M^{-1} L_1^\top \tilde{H} u_j \\ &= \frac{1}{2} u_j^\top \left[\tilde{H} - \tilde{H} L_1 M^{-1} L_1^\top \tilde{H} \right] u_j. \end{aligned} \quad (22)$$

This motivates the deflated Hessian

$$\tilde{H}_\perp = \tilde{H} - \tilde{H} L_1 \left(L_1^\top \tilde{H} L_1 \right)^{-1} L_1^\top \tilde{H}. \quad (23)$$

Hence,

$$\min_{\ell_j} \frac{1}{2} (u_j - L_1 \ell_j)^\top \tilde{H} (u_j - L_1 \ell_j) = \frac{1}{2} u_j^\top \tilde{H}_\perp u_j. \quad (24)$$

Moreover,

$$\tilde{H}_\perp L_1 = 0,$$

because

$$\begin{aligned}\tilde{H}_\perp L_1 &= \tilde{H} L_1 - \tilde{H} L_1 \left(L_1^\top \tilde{H} L_1 \right)^{-1} L_1^\top \tilde{H} L_1 \\ &= 0.\end{aligned}\tag{25}$$

Thus, the rank- r subspace spanned by L_1 lies in the null space of the residual quantization metric. Quantization errors along these directions can be absorbed by the high-precision low-rank branch.

Under the $\tilde{H}$ -orthogonal gauge

$$L_1^\top \tilde{H} L_1 = I,$$

the expressions simplify to

$$\boxed{\ell_j^* = L_1^\top \tilde{H} u_j}, \quad \boxed{\tilde{H}_\perp = \tilde{H} - \tilde{H} L_1 L_1^\top \tilde{H}}.$$

C PROTECTED-SUBSPACE ALLOCATION

The low-rank branch is not only a low-rank approximation of the weight matrix. Its input-side factor also determines a rank- r subspace in which quantization error can be absorbed by the high-precision branch. Let

$$\tilde{X} = X D^{-1}, \quad \tilde{W} = D W, \quad \tilde{H} = \tilde{X}^\top \tilde{X},$$

and write

$$\tilde{W} = L_1 L_2 + R, \quad L_1 \in \mathbb{R}^{c \times r}, \quad L_2 \in \mathbb{R}^{r \times n}.$$

The protected subspace is

$$\mathcal{S} := \text{range}(L_1) \subseteq \mathbb{R}^c.$$

For a fixed quantized residual column $\hat{w}_j$, eliminating the corresponding column ℓ_j of L_2 gives

$$\min_{\ell_j \in \mathbb{R}^r} \frac{1}{2} \left\| \tilde{X} (\tilde{w}_j - L_1 \ell_j - \hat{w}_j) \right\|_2^2 = \frac{1}{2} \left\| \tilde{w}_j - \hat{w}_j \right\|_{\tilde{H}_\perp}^2,$$

where

$$\tilde{H}_\perp = \tilde{H} - \tilde{H} L_1 \left(L_1^\top \tilde{H} L_1 \right)^{-1} L_1^\top \tilde{H}.$$

In particular,

$$\tilde{H}_\perp L_1 = 0.$$

Hence directions inside $\mathcal{S}$ are removed from the effective Hessian seen by the quantizer: errors along these directions are compensated by the high-precision low-rank branch. The rank budget r therefore determines how many directions can be protected, and the central question is how to allocate this budget.

C.1 WEIGHTED RANK- r APPROXIMATION

A natural starting point is to choose a rank- r matrix

$$L = L_1 L_2$$

that approximates $\tilde{W}$. Ordinary truncated SVD solves

$$\min_{\text{rank}(L) \leq r} \frac{1}{2} \left\| \tilde{W} - L \right\|_F^2,$$

which treats errors in all input directions equally. For layer reconstruction, however, the relevant quantity is the output perturbation

$$\tilde{X} (\tilde{W} - L).$$

This leads to the weighted problem

$$\min_{\text{rank}(L) \leq r} \frac{1}{2} \left\| \tilde{X} (\tilde{W} - L) \right\|_F^2.$$

Using

$$\tilde{H} = \tilde{X}^\top \tilde{X},$$

the objective can be written as

$$\frac{1}{2} \left\| \tilde{X}(\tilde{W} - L) \right\|_F^2 = \frac{1}{2} \text{tr} \left((\tilde{W} - L)^\top \tilde{H} (\tilde{W} - L) \right) = \frac{1}{2} \left\| \tilde{H}^{1/2} (\tilde{W} - L) \right\|_F^2.$$

Therefore, after the change of variable

$$M = \tilde{H}^{1/2} L,$$

the problem becomes an ordinary best rank- r approximation:

$$\min_{\text{rank}(M) \leq r} \frac{1}{2} \left\| \tilde{H}^{1/2} \tilde{W} - M \right\|_F^2.$$

By the Eckart–Young theorem,

$$M^* = \left[\tilde{H}^{1/2} \tilde{W} \right]_r,$$

and hence

$$L^* = \tilde{H}^{-1/2} \left[\tilde{H}^{1/2} \tilde{W} \right]_r.$$

Thus the weighted approximation protects directions that are simultaneously important in the weight matrix and strongly excited by the calibration activations.

C.2 HESSIAN EIGENSPACE LIMIT

The weighted approximation above is still a low-rank approximation problem. A second view is obtained by directly minimizing the expected quantization distortion after the protected subspace has been removed.

For output column j , let the weight quantization error be

$$\Delta w_j = w_j - \hat{w}_j.$$

Under an isotropic within-column quantization-noise model,

$$\mathbb{E} [\Delta w_j \Delta w_j^\top] = \frac{\delta_j^2}{12} I_c.$$

After eliminating L_2 , the expected distortion becomes

$$\mathbb{E} [\Delta w_j^\top \tilde{H}_\perp \Delta w_j] = \frac{\delta_j^2}{12} \text{tr} (\tilde{H}_\perp).$$

Summing over output columns gives

$$F_W \propto \text{tr} (\tilde{H}_\perp).$$

Let

$$\tilde{H} = V \Lambda V^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_c), \quad \lambda_1 \geq \dots \geq \lambda_c > 0.$$

The rank- r protected subspace that minimizes $\text{tr}(\tilde{H}_\perp)$ is the dominant eigenspace

$$\mathcal{S}^* = \text{span}\{v_1, \dots, v_r\}.$$

Under the $\tilde{H}$ -orthogonal gauge

$$L_1^\top \tilde{H} L_1 = I_r,$$

one convenient representative is

$$L_1^* = V_r \Lambda_r^{-1/2},$$

where $V_r = [v_1, \dots, v_r]$ and $\Lambda_r = \text{diag}(\lambda_1, \dots, \lambda_r)$.

This result gives a different interpretation from ordinary SVD. Plain SVD protects directions with large weight energy, whereas Hessian deflation protects directions in which a perturbation causes the largest layer-output distortion. When the rank budget is small, protecting these stiff directions can be more valuable than minimizing the unweighted residual norm.

C.3 BETA-FAMILY

The dynamic-range role of plain SVD and the sensitivity role of Hessian deflation can be interpolated by a one-parameter family. For $\beta \geq 0$, define

$$L^{(\beta)} = \tilde{H}^{-\beta} \left[\tilde{H}^{\beta} \tilde{W} \right]_r.$$

If

$$\tilde{H}^{\beta} \tilde{W} = U^{(\beta)} \Sigma^{(\beta)} V^{(\beta)\top},$$

then an associated input-side factor is

$$L_1^{(\beta)} = \tilde{H}^{-\beta} U_r^{(\beta)},$$

followed, if desired, by rescaling to satisfy

$$L_1^{(\beta)\top} \tilde{H} L_1^{(\beta)} = I_r.$$

The family includes two useful regimes, of which the main recipe uses $\beta = 1/2$.

For $\beta = 0$,

$$L^{(0)} = [\tilde{W}]_r,$$

which is the ordinary truncated-SVD branch. This endpoint retains the leading weight singular components.

For $\beta = \frac{1}{2}$,

$$L^{(1/2)} = \tilde{H}^{-1/2} \left[\tilde{H}^{1/2} \tilde{W} \right]_r,$$

which is exactly the weighted rank- r minimizer of

$$\min_{\text{rank}(L) \leq r} \frac{1}{2} \left\| \tilde{X}(\tilde{W} - L) \right\|_F^2.$$

The weight-aware family should not be identified with the Hessian-only trace-minimization problem above. In particular, the inverse factor $\tilde{H}^{-\beta}$ means that increasing β does not generally make the protected span converge to the leading Hessian eigenspace. The reported experiments fix $\beta = 1/2$ and do not use an adaptive activation-rank split or a global rank allocator.

D SMOOTHING OPTIMIZATION

The smoothing variables must be optimized while the low-rank branch is held in the original input coordinates. Write

$$L_1 = DA, \quad B = AL_2, \quad P = W - B.$$

When A and B are fixed, the smoothed residual is

$$R = \tilde{W} - L_1 L_2 = D(W - B) = DP.$$

Moreover, the deflated Hessian in these coordinates satisfies

$$\tilde{H}_{\perp} = D^{-1} H_{\perp} D^{-1}, \quad H_{\perp} = H - HA(A^{\top} HA)^{-1} A^{\top} H. \quad (26)$$

Thus $H_{\perp}$ and P remain fixed throughout the D -update. This coordinate choice also reflects the forward pass: the high-precision branch evaluates $\tilde{X} L_1 L_2 = X A L_2$ and does not itself need smoothing.

D.1 GROUPED ACTIVATION-NOISE MODEL

Partition the c input channels into activation-quantization groups $\mathcal{G}_A$. For calibration token t and group $G \in \mathcal{G}_A$, consider a symmetric b_A -bit quantizer with $\kappa_A = 2^{b_A-1} - 1$ and step size

$$\delta_{A,t,G}(d) = \frac{1}{\kappa_A} \max_{i \in G} \frac{|X_{ti}|}{d_i}.$$

Under the independent uniform-noise approximation, each channel in G has error variance $\delta_{A,t,G}^2/12$ at token t . Consequently, the grouped diagonal activation-noise covariance has entries

$$(\Sigma_A)_{ii} = \sigma_{G(i)}^2(d), \quad \sigma_G^2(d) = \frac{1}{12\kappa_A^2} \sum_{t=1}^m \max_{k \in G} \frac{X_{tk}^2}{d_k^2}. \quad (27)$$

Using $R = DP$, the pre-quantization activation-noise surrogate becomes

$$F_A^{\text{grp}}(d) = \frac{1}{12\kappa_A^2} \sum_{G \in \mathcal{G}_A} \left(\sum_{t=1}^m \max_{k \in G} \frac{X_{tk}^2}{d_k^2} \right) \left(\sum_{i \in G} d_i^2 \|P_{i,:}\|_2^2 \right). \quad (28)$$

The first factor measures the activation range seen by a group; the second measures how much residual weight that group carries. Their product explains why reducing an activation outlier is useful only insofar as the corresponding weight rescaling does not offset the gain.

For a singleton group $G = \{i\}$, d_i^{-2} in the first factor cancels d_i^2 in the second, so that group's contribution is independent of d_i . For groups containing multiple channels, the maximum can switch between channels while the residual energy changes continuously; smoothing then has a genuine within-group effect. The limiting case $G = \{1, \dots, c\}$ gives one activation scale per token.

The weight-noise channel has a parallel grouped model. Let $\mathcal{G}_W$ be the weight-quantization groups along the input dimension, and let $\kappa_W = 2^{b_W-1} - 1$. For output column j , define

$$\delta_{W,j,G}(d) = \frac{1}{\kappa_W} \max_{k \in G} d_k |P_{kj}|, \quad G \in \mathcal{G}_W.$$

With $h_i = (H_\perp)_{ii}$, the uniform-noise surrogate after eliminating L_2 is

$$F_W^{\text{grp}}(d) = \frac{1}{12} \sum_{j=1}^n \sum_{G \in \mathcal{G}_W} \delta_{W,j,G}^2(d) \sum_{i \in G} \frac{h_i}{d_i^2}. \quad (29)$$

Only the diagonal of $H_\perp$ appears here because the assumed weight errors are uncorrelated within a group. The off-diagonal entries remain relevant to the actual GPTQ solve below.

D.2 GEOMETRIC-PROGRAM FORMULATION

For fixed A and B , minimize the modeled two-channel objective

$$\min_{d_i > 0} F_W^{\text{grp}}(d) + \lambda_A F_A^{\text{grp}}(d), \quad \lambda_A \geq 0.$$

Introduce positive epigraph variables $t_{j,G}$ for weight steps and $s_{t,G}$ for activation steps. An equivalent geometric program is

$$\begin{aligned} \min_{d,t,s>0} \quad & \frac{1}{12} \sum_{j=1}^n \sum_{G \in \mathcal{G}_W} t_{j,G}^2 \sum_{i \in G} h_i d_i^{-2} \\ & + \frac{\lambda_A}{12} \sum_{G \in \mathcal{G}_A} \left(\sum_{t=1}^m s_{t,G}^2 \right) \left(\sum_{i \in G} d_i^2 \|P_{i,:}\|_2^2 \right) \\ \text{subject to} \quad & \frac{d_i |P_{ij}|}{\kappa_W t_{j,G}} \leq 1 \quad (i \in G), \\ & \frac{|X_{ti}|}{\kappa_A d_i s_{t,G}} \leq 1 \quad (i \in G), \\ & \prod_{i=1}^c d_i = 1. \end{aligned} \quad (30)$$

The first constraint applies to every output j and weight group $G \in \mathcal{G}_W$; the second applies to every token t and activation group $G \in \mathcal{G}_A$. At an optimum, each epigraph variable can be reduced to its corresponding maximum, recovering (28) and (29). All objective terms are posynomials with nonnegative coefficients, the inequalities are monomial bounds, and the normalization is a monomial equality. Hence the change of variables

$$u_i = \log d_i, \quad \tau_{j,G} = \log t_{j,G}, \quad \xi_{t,G} = \log s_{t,G}$$

turns the constraints into affine inequalities and the logarithm of the objective into a convex log-sum-exp function. The idealized fixed-branch D -subproblem therefore has a global optimum, subject to the stated noise model and any imposed bounds on d .

The normalization removes an otherwise arbitrary common scaling: replacing every d_i by ad_i leaves both modeled error terms unchanged. Equivalently, one may impose $\sum_i u_i = 0$. If a practical implementation clips the smoothing range, bounds $-\log d_{\max} \leq u_i \leq \log d_{\max}$ can be added without losing convexity. A finite number of gradient steps in log space is an approximate solver for this geometric program; the global-optimum claim applies to a converged solution of the modeled subproblem, not automatically to a finite-step implementation or to the measured quantized-layer error.

D.3 CLOSED-FORM RELAXATION

A simpler initialization follows by replacing grouped maxima with an ungrouped ℓ_p range surrogate. Define

$$\omega_i^p = \sum_{j=1}^n |P_{ij}|^p, \quad h_i = (H_\perp)_{ii}, \quad p \geq 1,$$

and consider

$$\hat{F}_W(d) = C \left(\sum_{i=1}^c \omega_i^p d_i^p \right)^{2/p} \left(\sum_{i=1}^c h_i d_i^{-2} \right), \quad (31)$$

where $C > 0$ does not affect the minimizer. Assuming $h_i > 0$ and $\omega_i > 0$, differentiation of $\log \hat{F}_W$ with respect to $u_i = \log d_i$ gives

$$\frac{2\omega_i^p d_i^p}{\sum_k \omega_k^p d_k^p} - \frac{2h_i d_i^{-2}}{\sum_k h_k d_k^{-2}} = 0.$$

The ratio of the two denominators is common to every channel. Therefore, up to an overall scale,

$$d_i^* \propto \left(\frac{h_i}{\omega_i^p} \right)^{1/(p+2)}. \quad (32)$$

Because the logarithm of (31) is convex in u , this stationary point is a global minimizer of the relaxation. Normalize its geometric mean to one before refinement.

For $p = 2$,

$$d_i^* \propto \frac{h_i^{1/4}}{\|P_{i,:}\|_2^{1/2}}.$$

If the low-rank branch is absent, then $h_i = H_{ii} = \|X_{:,i}\|_2^2$ and $P = W$, yielding

$$d_i^* \propto \left(\frac{\|X_{:,i}\|_2}{\|W_{i,:}\|_2} \right)^{1/2}.$$

The square-root rule is thus exact for this relaxation, while the grouped maxima and activation-noise term motivate the subsequent refinement.

D.4 THE ACTIVATION-AWARE METRIC AND TARGET

For fixed D, L_1 , the modeled activation penalty depends on $\hat{R}$ but not L_2 . Eliminating L_2 therefore gives

$$J(\hat{R}) = \frac{1}{2} \|\widetilde{W} - \hat{R}\|_{H_\perp}^2 + \frac{1}{2} \lambda_A \|\hat{R}\|_{\Sigma_A}^2.$$

Define $M = H_\perp + \lambda_A \Sigma_A$. Both summands are positive semidefinite, implying $\text{null}(M) \subseteq \text{null}(H_\perp)$. Consequently $H_\perp \widetilde{W} \in \text{range}(M)$ and $T = M^\dagger H_\perp \widetilde{W}$ satisfies $MT = H_\perp \widetilde{W}$. Expanding the square yields

$$J(\hat{R}) = \frac{1}{2} \|\hat{R} - T\|_M^2 + \frac{1}{2} \|\widetilde{W}\|_{H_\perp}^2 - \frac{1}{2} \|T\|_M^2.$$

Thus both the target and metric change when activation noise is included. At $\lambda_A = 0$, the metric is generally singular: an ordinary inverse cannot replace the pseudoinverse without regularization.

What regularization changes. With a common loaded metric $M_\epsilon = M + \epsilon I$ and target $T_\epsilon = M_\epsilon^{-1} H_\perp \tilde{W}$, the corresponding objective is $J(\hat{R}) + \frac{1}{2}\epsilon \|\hat{R}\|_F^2$, up to a code-independent constant. The weight-error path instead quantizes an initialized residual $R_0 = \tilde{W} - L_1 L_{2,0}$ under a stabilized $H_\perp$. Its added penalty is centered at R_0 . In addition, group scales are constructed from the chosen target. Consequently, equal continuous undamped losses need not produce equal grids, greedy updates, or codes.

Exact activation error versus its surrogate. The exact error is $\tilde{X}C - E_A \hat{R}$ and includes the cross term in Eq. (5). Replacing deterministic activation rounding with a zero-mean conditional noise model is the step that removes this cross term in expectation. The deployed activation error multiplies $\hat{R}$, not the unquantized provisional residual. Our post-code objective preserves that distinction. The diagonal range-based covariance does not estimate off-diagonal correlations and is not an exact model of four-bit rounding.

E ALGORITHM

Algorithm 1 Detailed layerwise calibration of H -SVDQuant

Require: Weight W , calibration activations X , protected rank r , symmetric INT4 weight quantizer Q_W , per-token group-wise INT4 activation quantizer Q_A , code mode, activation weight λ_A

Ensure: $D, L_1, L_2, \hat{R}$

```

1:  $H_0 \leftarrow X^\top X / m$ 
2: Initialize best candidate state
3:  $A_{\text{prev}} \leftarrow \emptyset, L_{2,\text{prev}} \leftarrow \emptyset$ 
4: for  $k = 1, 2$  do
5:   if  $k = 1$  then
6:      $H_{\perp,\text{prev}} \leftarrow H_0$ 
7:      $P_{\text{prev}} \leftarrow W$ 
8:   else
9:      $H_{\perp,\text{prev}} \leftarrow H_0 - H_0 A_{\text{prev}} (A_{\text{prev}}^\top H_0 A_{\text{prev}})^\dagger A_{\text{prev}}^\top H_0$ 
10:     $P_{\text{prev}} \leftarrow W - A_{\text{prev}} L_{2,\text{prev}}$ 
11:   end if
12:   Update  $D$  from  $(H_{\perp,\text{prev}}, P_{\text{prev}}, X)$  using Equation (13) and optional cached refinement
13:    $\tilde{X} \leftarrow X D^{-1}, \tilde{W} \leftarrow D W, \tilde{H} \leftarrow D^{-1} H_0 D^{-1}$ 
14:   Estimate  $\bar{\Sigma}_A$  using Equation (6)
15:   Initialize  $(L_1, L_{2,0})$  by the weighted rank- $r$  fit in Equation (12)
16:   Form the deflated metric  $\tilde{H}_\perp$  using Equation (9)
17:   if activation-aware code optimization is enabled then
18:     Form  $(M_Q, T_Q)$  using Equation (11)
19:     Construct  $\hat{R}$  by GPTQ on target  $T_Q$  with metric  $M_Q$  using  $Q_W$ 
20:   else
21:      $R_0 \leftarrow \tilde{W} - L_1 L_{2,0}$ 
22:     Construct  $\hat{R}$  by GPTQ on target  $R_0$  with metric  $\tilde{H}_\perp$  using  $Q_W$ 
23:   end if
24:   Refit  $L_2$  using Equation (8)
25:   Evaluate the W4A4 output MSE on cached calibration inputs in FP32 and retain the best visited state
26:    $A_{\text{prev}} \leftarrow D^{-1} L_1, L_{2,\text{prev}} \leftarrow L_2$ 
27: end for
28: return best  $(D, L_1, L_2, \hat{R})$ 

```

F EVALUATION ON ACCELERATED DIFFUSION BACKBONES

Table 8 reports quantitative W4A4 results on accelerated few-step diffusion backbones: the 4-step FLUX.1-schnell and 4-step Qwen-Image Lightning variants, complementing the standard base model evaluations in Table 1. Teacher fidelity and standalone quality measure different properties: the method with the best PSNR need not have the highest CLIP or ImageReward score. Thus these evaluations capture distinct trade-offs between teacher fidelity and generation metrics across few-step regimes.

Table 8: **W4A4 fidelity and generation quality on accelerated diffusion backbones (FLUX.1-schnell and Qwen-Image Lightning)**. Both models use 4-step sampling on the prescribed dataset. CLIP and ImageReward (IR) are absolute scorer means, not teacher-relative differences; neither measures teacher fidelity. Bold and underline mark the best and second-best unrounded means per subset and metric. *H-SVDQuant* use group-64 residuals and the stated protected rank; DiRotQ uses matched A4/group-64.

Method	MJHQ				sDCI			
	PSNR $\uparrow$	LPIPS $\downarrow$	CLIP $\uparrow$	IR $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	CLIP $\uparrow$	IR $\uparrow$
FLUX.1-schnell (4 steps)								
H-SVDQuant $r = 4$	<u>17.93</u>	<u>0.260</u>	<u>25.238</u>	0.9029	<u>16.79</u>	<u>0.278</u>	<u>25.838</u>	1.0761
SVDQuant $r = 4$	16.18	0.350	25.394	0.8951	15.12	0.363	25.791	1.0844
SVDQuant $r = 32$	17.31	0.298	25.223	0.9394	16.23	0.311	25.712	<u>1.0947</u>
DiRotQ	18.56	0.241	24.994	<u>0.9089</u>	17.41	0.257	25.588	1.0951
OrbitQuant	17.21	0.297	25.144	0.9007	16.23	0.306	25.836	1.0413
ViDiT-Q	14.74	0.467	24.758	0.7872	13.85	0.489	26.142	1.0235
Qwen-Image Lightning (4 steps)								
H-SVDQuant $r = 4$	15.33	<u>0.328</u>	29.422	1.6268	<u>13.33</u>	<u>0.391</u>	25.541	1.2425
SVDQuant $r = 4$	14.31	0.382	29.639	1.5958	12.84	0.437	25.508	1.2173
SVDQuant $r = 32$	15.09	0.339	29.580	1.6063	13.30	0.406	25.602	1.2372
DiRotQ	<u>15.18</u>	0.327	<u>29.597</u>	<u>1.6160</u>	13.50	0.383	<u>25.543</u>	<u>1.2400</u>

G QUALITATIVE RESULTS OF QWEN-IMAGE

Figures 3 and 4 show representative qualitative reconstructions for the 50-step base model and four-step Lightning variant, respectively, evaluated under identical seeds and prompts against DiRotQ and both SVDQuant ranks.

Qwen-Image. A group of children are sitting in the back of a convertible car. There are buildings and vehicles in the background.

FP16/BF16

H-SVDQuant

DiRotQ

SVDQuant $r = 4$

SVDQuant $r = 32$



Qwen-Image. black mamba up close

FP16/BF16

H-SVDQuant

DiRotQ

SVDQuant $r = 4$

SVDQuant $r = 32$



Qwen-Image. This shows a somewhat dilapidated looking shelter in front of a white gazebo. There is a sloped hill in the background.

FP16/BF16

H-SVDQuant

DiRotQ

SVDQuant $r = 4$

SVDQuant $r = 32$



Qwen-Image. A curved staircase encased in glass is outside of a building.

FP16/BF16

H-SVDQuant

DiRotQ

SVDQuant $r = 4$

SVDQuant $r = 32$

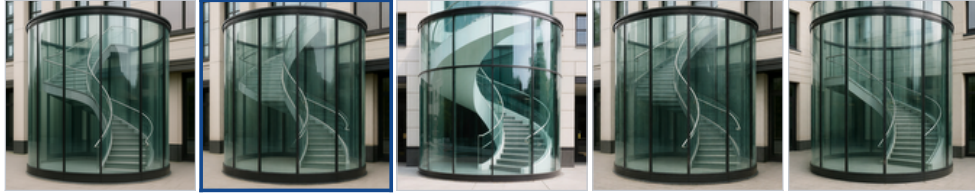
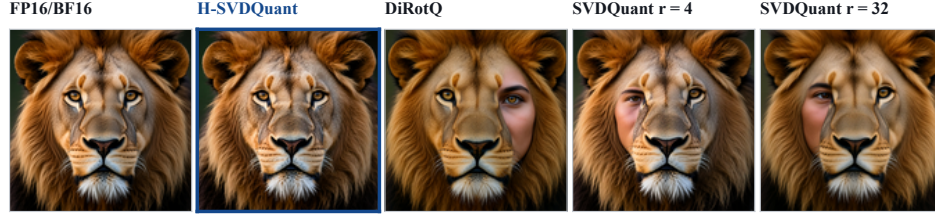
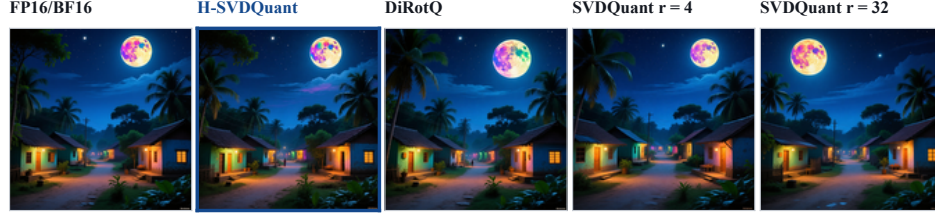


Figure 3: **Qwen-Image base-model reconstructions.** The first column is the high-precision reference; the blue outline identifies *H-SVDQuant*. All methods share the displayed prompt within each row. These qualitative examples complement the complete-subset results in Table 1.

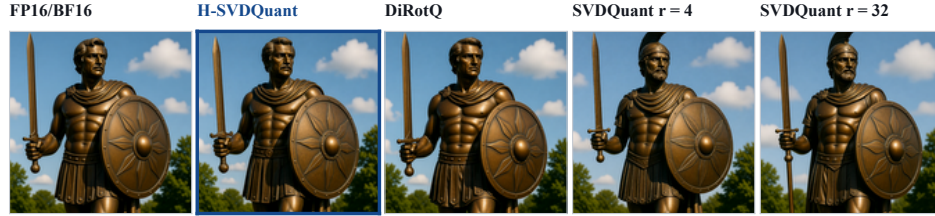
Qwen-Image Lightning. An anthropomorphic lion face. The face is a mix of a human and a lion face. Award winning photography. Closeup



Qwen-Image Lightning. Design a photorealistic village night landscape set in Bangladesh in 1970s, inspired by Bangladeshi artist S.M.Sultan. The image should feature bright and colorful full moon night,



Qwen-Image Lightning. There is a bronze statue of a middle aged man dressed as an ancient warrior who has a sword at his side as well as a large shield. The sky and trees are visible in the background.



Qwen-Image Lightning. A passenger airplane is captured just as it is taking off. On the side of the airplane is written the company name in lowercase: "indonesian airways". The airplane's front wheel is in the air, and its back wheels are just



Qwen-Image Lightning. A long, huge brown roof made of tiles with four chimneys on its top. To the left of the roof is ocean with a seagull flying above.

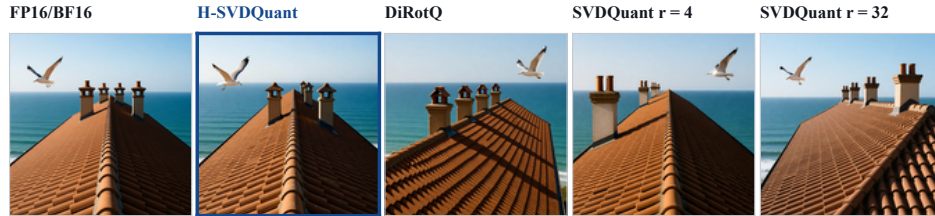


Figure 4: **Four-step Qwen-Image Lightning reconstructions.** Column order follows Figure 3. The accelerated quantitative comparison appears in Table 8.

H FULL PAIRED PROMPT DISTRIBUTIONS

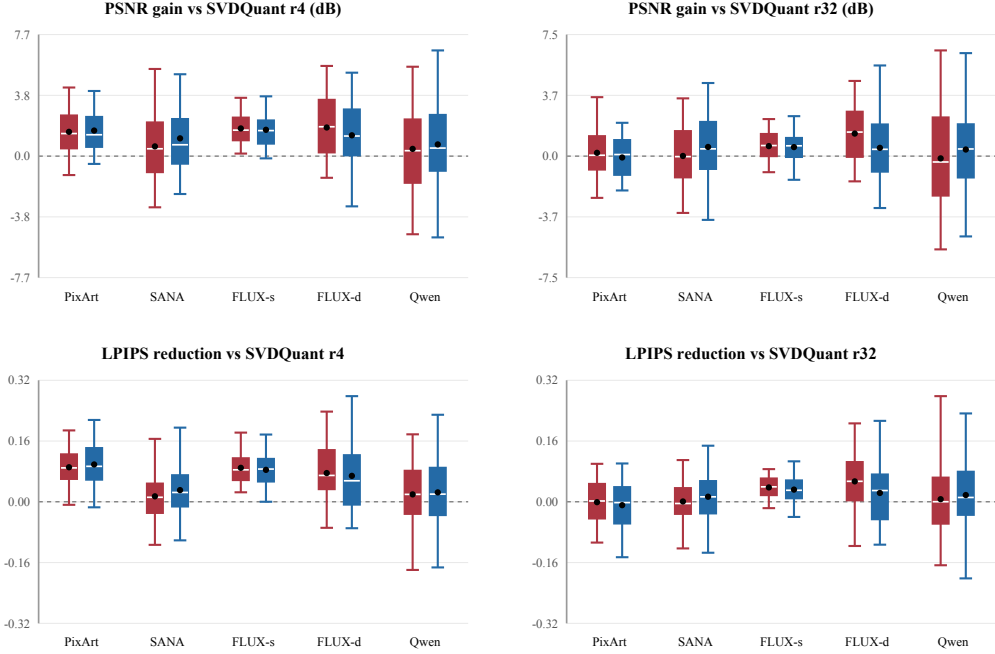


Figure 5: **Paired teacher-fidelity differences across five diffusion backbones.** This expanded version of Figure 1 adds Qwen-Image and compares rank-4 *H-SVDQuant* against SVDQuant ranks 4 and 32 on identical prompts and seeds. Columns separate baseline rank; rows show PSNR gain and LPIPS reduction. Boxes show the interquartile range, white bars the medians, whiskers the 5th–95th percentiles, and black dots the means. Red denotes MJHQ and blue sDCI; positive values favor *H-SVDQuant*. The compared checkpoints retain their respective activation quantizers and are complete-recipe comparisons. Qwen-Image images are paired by index under identical prompt, seed, and teacher protocols.

I BETWEEN-BLOCK AND WITHIN-BLOCK PROPAGATION PILOT

Table 9: **Three-way calibration replay pilot.** All rows use rank 4, W4A4/group-64, $\lambda_A = 0$, and the weight-error code path. Within each model, evaluation uses the same 16 MJHQ prompts and image identifiers. These pilot checkpoints are not substituted into Table 1.

Model	Between blocks	Within block	PSNR	LPIPS
PixArt- Σ	quantized	sequential	15.523	0.4834
	reference	sequential	15.701	0.4675
	reference	independent	15.833	0.4662
FLUX-schnell	quantized	sequential	17.471	0.3003
	reference	sequential	17.703	0.2961
	reference	independent	17.059	0.3151

Relative to reference/sequential, the two PixArt PSNR differences are -0.177 dB (95% paired interval $[-0.839, 0.412]$) and $+0.133$ dB ($[-0.239, 0.529]$). This pilot does not establish a ranking. For FLUX-schnell, the corresponding differences are -0.232 dB ($[-0.500, 0.016]$) and -0.644 dB ($[-1.395, -0.003]$); both LPIPS intervals include zero. Despite worse image fidelity in the independent FLUX run, its largest local module MSE is slightly smaller. Local errors are measured on each route’s own calibration activations and are not paired errors on a shared held-out activation set. The pilot therefore illustrates a limitation of local MSE as a predictor of full-image behavior.